

Practise Questions

241-2516 BDS-4B

Q1)

- (b) Testing under ^{vector} addition closure
set $W = (a, 1, 1)$ form

$$(a, 1, 1) + (b, 1, 1) = (a+b, 2, 2)$$

resultant does not belong to W

hence W is not a subspace of $\mathbb{R}^3$

Q2)

- (c) Testing under vector addition closure

set $W = (a, b, c)$ where $a+b=7$

$$(a, b, c) + (a', b', c') = (a+a', b+b', c+c')$$

if $a+b=7$ and $a'+b'=7$

then resultant's sum $a+a' + b+b' = 14$

is not belonging to W .

as resultant does not belong to W it fails
vector addition and $\vec{0}$ so W is not subset of
 $\mathbb{R}^3$.

Q2)

(a) $W = n \times n$ diagonal matrix

W contains at least one matrix (the zero $n \times n$ matrix)
checking ^{vector} scalar addition closure

$$\begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & & \\ \vdots & & \ddots & \\ 0 & 0 & \dots & a_{nn} \end{bmatrix} + \begin{bmatrix} a'_{11} & 0 & \dots & 0 \\ 0 & a'_{22} & & \\ \vdots & & \ddots & \\ 0 & 0 & \dots & a'_{nn} \end{bmatrix} = \begin{bmatrix} a_{11}+a'_{11} & 0 & \dots & 0 \\ 0 & a_{22}+a'_{22} & & \\ \vdots & & \ddots & \\ 0 & 0 & \dots & a_{nn}+a'_{nn} \end{bmatrix}$$

resultant belongs to W .

Testing scalar multiplication closure

$$k \begin{bmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{22} & & \\ \vdots & & \ddots & \\ 0 & 0 & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} ka_{11} & 0 & \dots & 0 \\ 0 & ka_{22} & & \\ \vdots & & \ddots & \\ 0 & 0 & \dots & ka_{nn} \end{bmatrix}$$

resultant belongs to W .

As it passes all tests, W is a subspace of M_{nn}

Q7)

(a) ~~Q2~~ all functions f in $F(-\infty, \infty)$ for which $f(0) = 0$

Check no 1: Zero vector

it is contained as zero function $0(x) = 0$ exists

Check no 2: Scalar Addition Closure

if $f(0) = 0$ and $g(0) = 0$, then $h(x) = g(x) + f(x)$ will be

$$h(0) = 0 + 0 = 0.$$

This satisfies condition.

Check no. 3 = scalar Multiplication closure (assume $f(0) = 0$)
if $g(x) = c \cdot f(x)$, then $g(0) = c \cdot 0 = 0$
This satisfies the condition

as all conditions are satisfied f is a subspace
under $F(-\infty, \infty)$.

Q7)

(b) f is all functions in $F(-\infty, \infty)$ for which $f(0) = 1$

Checking closure under addition

$$\text{if } f(0) = 1, g(0) = 1,$$

$$\text{then for } h(x) = f(x) + g(x)$$

$$\text{result for } h(0) = 1 + 1 = 2$$

This result is not in f so no closure.

Thus f is not a subspace under $F(-\infty, \infty)$.

Q9)

(a) $S =$ all sequences in $\mathbb{R}^\infty$ in form $(v, 0, v, 0, \dots)$ ~~any~~
real number

Check no. 1 = Zero / Empty check

sequence $(0, 0, 0, \dots)$ is present hence pass.

Check no. 2 = Addition closure

$$\text{For 2 sequences } (v, 0, v, 0, \dots), (v', 0, v', 0, \dots)$$

$$(v, 0, v, 0, \dots) + (v', 0, v', 0, \dots) = (v+v', 0, v+v', 0, \dots)$$

resultant is in S so there is closure

Check no. 3: Scalar Multiplication closure

$$k(v, 0, v, 0, \dots) = (kv, 0, kv, 0, \dots)$$

resultant is in S so there is closure

All checks passed so ^S is a subspace of $\mathbb{R}^\infty$

(b) $S =$ all sequences in $\mathbb{R}^\infty$ in form $(v, 1, v, 1, \dots)$

check addition closure

$$(v, 1, v, 1, \dots) + (v', 1, v', 1, \dots) = (v+v', 2, v+v', 2, \dots)$$

resultant is not in S so no closure

hence S is not a subspace of $\mathbb{R}^\infty$.

Q16)

(a) $W =$ set of all polynomials with even coefficients

Testing closure under scalar multiplication

e.g. $P(x) = 6x$, $c = \frac{1}{2}$

$$c \cdot p(x) = \frac{6}{2}x = 3x$$

resultant is not in W so no closure

W is not a subspace of $\mathbb{R}^\infty P_\infty$